

A Procedural Production System for Autonomous Vehicle Simulation

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Figure 1: Examples of virtual worlds built using our production system. Left: High fidelity models are combined with procedural content creation based on HD map data. Right: Lidar point cloud view of a virtual scene, showing return intensity in the color channel.

ABSTRACT

Simulations provide the only safe and reliable way to test autonomous vehicle systems in situations that may be unusual or undesirable to witness on the road. In order to make simulation at scale possible and its accuracy sufficient, production systems must be built which adapt state-of-the-art graphics production practices to this new field. In this work we discuss the procedurally based system developed at Aurora, which forms a key part of the company's approach to testing and validation.

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1 INTRODUCTION

Simulation plays a key role in both the development and testing of autonomous vehicles, with safety and repeatability being two notable advantages over on-road and track testing. As performance of an autonomy system is successfully validated for nominal situations using a combination of real-world and virtual tests, simulations provide the only practical way to then extend the validation to situations which are unusual to observe, or inherently unsafe to enact. One particularly challenging aspect of this testing is the perception subsystem, which consumes multiple modalities of sensor data (cameras, lidar, radar) and outputs semantic information about the vehicle's surroundings. Simulators that provide sensor data must be highly realistic and simulated scenarios must provide a large degree of variability, in order to provide coverage as well as confidence in the simulation results.

In this work, we discuss a production system which utilizes technologies (e.g., OpenUSD) and processes (e.g., storyboarding, animation previews) from visual effects and animation, but which is built expressly for the purpose of allowing a small team of TDs and engineers to produce high fidelity sensor-enabled simulations

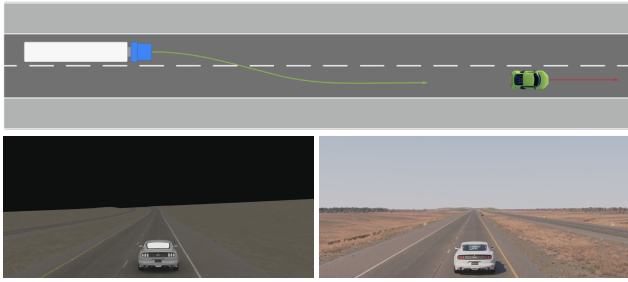


Figure 2: Examples of scenario development phases. Top: Diagram outlining the interaction of interest. Bottom left: Animation preview (one instance out of a large set.) Bottom right: Final scenario (one instance out of a large set.)

for autonomous vehicle testing and validation. We highlight the procedural aspects of the system, which at peak times generates the equivalent of hundreds of feature films and petabytes of data on a weekly basis. Additionally, we discuss the processes and dynamics of a simulation team working with partner autonomy teams within the company, a collaboration which is not unlike that found in animation projects where the director and the production crew all operate within the same organization.

2 PRODUCTION PROCESS

When using simulations in autonomous vehicle development, it is important to consider that all data must be good data. There is little value in entirely random generation of driving situations; rather, there needs to be a purpose behind each situation created, whether used to train or test the system. To that end, our production process focuses on the delivery of large numbers of simulations, tailored toward the evaluation of specific capabilities of the autonomy system, and whose scope is defined in a collaborative process between the simulation, autonomy, and safety teams.

While there are clear parallels to the film making process, there are also key differences when building a synthetic data pipeline. In the film context, TDs and artists are iterating towards achieving a director’s singular vision of a particular shot. In the context of synthetic data, the goal is instead to create a wide range of related variations of a particular scenario, covering all aspects that may influence that particular situation. To that end, our process aims to enable TDs to generate, review, and collaborate on many hundreds of variations of any given test scenario concurrently, in a process that starts from high-level diagrams (essentially storyboards), progresses into situational visualizations (animation previews) and eventually turns into full-fledged sensor simulations (final render), as illustrated in Figure 2. Throughout the process, our simulated data is presented to the autonomy system through all the same interfaces that the physical vehicle uses, ensuring that the virtual world carefully mimics the real.

3 PIPELINE AND PRODUCTION WORKFLOW

The core of our scenario creation pipeline is built around Houdini and OpenUSD. In Houdini we use custom Digital Assets to build a fully synthetic environment based on HD maps as well as on-road

sensor data, and we utilize a wide range of procedural processes in order to create the variations desired for any given scenario. Our system includes tooling for TDs to construct parameterized road interactions, such as cut-ins, collisions, and near-collisions - all situations that would be uncomfortable or unsafe to test physically. Direct simulation of the sensor data also enables controlled repeatability and variability which are important for validating changes to the system.

Once created, the USD scene is then passed to our custom sensor simulation (i.e., rendering) engine, which acts as a virtual stand-in for the physical devices found on our vehicles. For details on the sensor simulation process, see Villemain et al. [Villemain et al. 2024]. As our autonomy system is multimodal, using cameras as well as lidar and radar, the assets used in the simulation system must be created with this in mind, and we employ a custom USD material schema that unifies not only spectral visible light properties, but also considers near-infrared and millimeter radar frequencies, with a stack of layered BRDFs for each. Our system also utilizes USD’s compositional capabilities in order to add and manipulate ML labeling and metadata information on each asset.

To generate the petabytes of data that a test set consists of on short timelines, we built our compute workflow around an AWS-based OpenCue deployment with containerized task execution, enabling us to scale capacity from zero to hundreds of thousands of CPUs as needed. Taking a cloud-first approach has been key to making this scaling possible, and our system implements file caching strategies that leverage common block storage, rather than the distributed file systems that graphics production pipelines typically utilize.

4 WORKING WITH PARTNER TEAMS

Once sensor simulations are complete, our system provides tools for efficiently reviewing both the large number of resulting test scenarios, as well as the autonomy system’s behavior in each. This review process is highly iterative and is designed to give all stakeholders visibility on the available data, such that insights about both test coverage and autonomy performance can be turned into refinements of the test configuration. This process is analogous to the dailies concept in film production, and has been key to achieving the simulation quality needed to be able to trust the results for validation purposes.

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REFERENCES

Ryusuke Villemain, Magnus Wrenninge, Steve Capell, Anton Gribovskiy, I-Chen Jwo, and Steve Bako. 2024. Alba: A Multimodal Rendering System for Autonomous Vehicle Simulation. In *ACM SIGGRAPH 2024 Talks*.